MOUAD IMAM

Full-Stack Engineer (Ruby on Rails & React) | PhD Candidate, Machine Learning

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Full-stack engineer with strong Ruby on Rails expertise, building production features in Rails and React. Led a team of 3 as Tech Lead, making the architecture decisions and running code reviews while staying hands-on across the stack: Sidekiq automation, Hotwire/Stimulus, ActionCable, platform security (JWT, CORS, Kubernetes hardening), and performance tuning (N+1 elimination, caching). Delivered across enterprise ERP, medical imaging, and a French B2B accounting SaaS reaching 3,200+ clients. Also a PhD candidate applying reinforcement learning to real-time routing in urban transit.

Experience

Myunisoft (GeFi) - French accounting SaaS — Remote, France
Full-Stack Engineer — Jan. 2026 - Present

- Acted as the cross-team liaison on a multi-team initiative, serving as single point of contact for inbound issues and driving fixes across teams to unblock partner delivery.
- Contributed to GeFi's addressing-identifier feature for France's RFE (Réforme Électronique) e-invoicing rollout, reaching 3,200+ clients on a French accounting SaaS.
- Conducted a security audit across the SaaS and its Kubernetes cluster, shipping fixes to harden authentication, network policies, and service exposure.
- Authored tenant-wide background jobs that automate business-contact enrichment from public company registries and migrate legacy data across all tenants, eliminating recurring manual ops.
- Resolved systemic performance bottlenecks across the admin console, significantly cutting load times on workflows operations teams depend on daily.

Timlead - Enterprise resource planning — Morocco
Tech Lead — Jul. 2024 - Dec. 2025

- Led a team of 3 engineers as Tech Lead: made the architecture decisions, ran code reviews, and mentored the team on a platform that grew to 5,000+ field agents and up to 60,000 tickets.
- Built a template-driven ticketing platform with workflows and conditional logic that cut template rollout time by ~50% while keeping field layouts consistent across projects and clients.
- Delivered a React/Stimulus response editor with Word/PDF exports and entity mapping so agents capture structured data once and reuse it across web, API, and document outputs.
- Implemented review and auto-review states with ActionCable notifications, map marker coloring, and SLA-aware visibility so ops teams see status changes as they happen.
- Shipped offline-ready bundles that inline media and signatures, bypass conditional blockers safely, and keep validations consistent, letting field agents stay productive without connectivity.
- Migrated the stack to Dockerized CI/CD with bin/dev hot reloads and overcommit hooks, shrinking builds from 15 to 4 minutes and smoothing steady Turbo/React delivery.
- Built CSV/XLSX import and export with column analysis, mapping validation, and error surfacing, reducing support escalations during bulk ticket creation and reporting.
- Automated dispatch scheduling using distance and workload scoring, added calendar and trajectory views, and supported 5,000+ field agents with assignments and 99% uptime.
- Tightened permissions and template lifecycle controls—clone flows, site-scoped policies, attachment validation—so admins ship new workflows safely without exposing sensitive fields.
- Instrumented ticket analytics for state counts, completion rates, and dispatch time, giving ops real-time signals and faster SLA triage.
- Added offline JSON payloads for responses, inlining blobs so agents sync cleanly after reconnecting and preserving field validations.

Radiology Information System (RIS) - Mibtech — Morocco
Software Developer — Feb. 2025 - May. 2025

- Integrated Orthanc as a PACS server and configured DICOM Modality Worklist (MWL) to automate exam scheduling from the RIS
- Designed and optimized the database schema (PostgreSQL) for patients, modalities, imaging studies, rooms, and medical reports ensuring high data consistency and traceability
- Implemented user role-based access control (RBAC) to secure patient data and enforce medical confidentiality regulations
- Developed API endpoints to enable real-time synchronization between RIS and PACS systems using DICOM networking protocols (C-FIND, C-STORE)
- Created intuitive user interfaces for patient registration, exam ordering, and report validation to improve workflow efficiency for radiology staff
- Monitored system logs and DICOM communication events to detect, troubleshoot, and resolve integration errors between imaging devices and the platform
- Wrote detailed technical documentation including workflow diagrams, deployment guidelines, and integration specifications for long-term maintainability
- Collaborated with DevOps practices by containerizing RIS/PACS components using Docker and configuring services for reliable on-premise deployment

Humaneo - Mibtech Morocco
Software Developer Aug. 2023 - May. 2025

- Added constraint-based rules to the employee scheduling module to avoid overlapping shifts, night-shift limits, and contractual violations
- Reduced CI build duration by improving Docker layer caching and splitting unit/integration tests, which helped speed up deployments
- Built notification emails for approvals, pending actions, and onboarding steps to make sure HR and managers respond on time
- Developed a simple internal training space where employees can watch videos, answer quizzes, and track their completion status
- Refactored parts of the employee management module to simplify data updates and avoid duplicate information across departments
- Worked closely with HR staff to adapt rules (leave approvals, contract renewal alerts…) directly into the workflows
- Improved system reliability by adding input validations and clearer error handling on critical forms (contracts, leave requests…)
- Wrote technical notes and usage guidelines so that onboarding new devs and HR users became easier

Projects

ProdLoger - Intelligent Log Management Platform Morocco
Academic Project - Master's Thesis (FSAC) Feb. 2025 - Jul. 2025

- Developed a real-time log monitoring and anomaly detection system powered by a BiLSTM deep learning model
- Handled ≈120K logs/sec with 82.76% precision and 93.31% recall during performance tests
- Implemented core services in Rust to improve throughput and reduce inference latency
- Used Elasticsearch for indexed log storage and PostgreSQL for persistent metadata
- Designed a preprocessing pipeline and tuned thresholds to reduce false positives and improve alert consistency
- Built dashboards in Grafana and collected metrics with Prometheus to monitor latency, throughput, and alert activity
- Containerized components (collector, inference engine, monitoring) with Docker for easier scaling
- Automated load tests with GitHub Actions to detect performance regressions before deployment
- Proposed future improvements including Transformer-based models and automated hyperparameter tuning

Mangel - Online Reading Platform Morocco
Personal Project Feb. 2024 - Jun. 2024

- Built a web application for browsing and reading manga/novels using a Ruby on Rails backend and a React front-end
- Designed REST API endpoints for chapters, reading progress, user libraries, and content discovery
- Handled image and chapter delivery with optimized pagination to ensure smooth reading experience on web
- Implemented user authentication and favorites management so users can track their ongoing series
- Stored manga metadata and chapter files in PostgreSQL with structured indexing to speed up queries
- Deployed the platform on Kubernetes and configured secrets and environment keys for secure runtime setup
- Used Docker to containerize both backend and frontend for consistent local and cloud environments
- Added loading states, infinite scrolling, and prefetching logic in React to improve perceived performance
- Set up basic monitoring and logs to track errors and performance for future improvements

Skills

Programming Languages: Ruby, JavaScript/TypeScript, Python, PHP, Rust, SQL, Bash
Frontend: React, Stimulus.js, Next.js, Tailwind CSS, Hotwire, Axios
Backend & APIs: Ruby on Rails, Node.js, Express, NestJS, Laravel, Sidekiq, WebSockets, REST, JWT
Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Firebase, Elasticsearch
Cloud & DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Google Cloud, AWS, Vercel, Dokku
Monitoring & Observability: Sentry, Prometheus, Grafana, ClickHouse, Amplitude
Testing & QA: RSpec, Capybara, Jest, Cypress, Postman
Data & AI: TensorFlow/PyTorch (NLP), Model Training & Evaluation, Vector Search
Medical Tech Standards: DICOM, PACS, Orthanc, Modality Worklist (MWL)
Tools & Collaboration: Git, Docker Compose, Figma, Trello, Jira, Notion, Linux

Education

University of Hassan II, Casablanca Morocco
PhD candidate in Machine Learning (Reinforcement Learning & Multi-Agent Systems) 2025 - Present

Research: multi-agent reinforcement learning for real-time passenger routing in multimodal urban transit (bus, tram, BRT, taxi).

University of Hassan II, Casablanca Morocco
Master's in Big Data & Cloud Computing 2023 - 2025

University of Hassan II, Casablanca Morocco
Bachelor's in Computer Science 2020 - 2023